week 3

Shreshta Bathini

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Report: BERT Fine-tuning and Transformer-based Inference

Objective

The objective of this experiment was to apply transformer-based models to text classification tasks. The focus was on fine-tuning a pre-trained BERT model, training a custom subword tokenizer using SentencePiece, and performing inference using the Hugging Face Transformers library.

Methodology

A BERT model (`bert-base-uncased`) was fine-tuned on the SST-2 sentiment analysis dataset. The text data was tokenized using the BERT tokenizer and prepared using attention masks and padding. The Hugging Face `Trainer` API was used to simplify the training and evaluation process. In addition, a custom subword tokenizer was trained using the SentencePiece library on a small text corpus, demonstrating language-independent tokenization and subword segmentation.

For inference, the Hugging Face `pipeline` API was used to perform sentiment prediction on unseen text efficiently, enabling quick deployment of transformer models.

Conclusion

The experiment demonstrated the effectiveness of transformer-based architectures for text classification and highlighted the importance of subword tokenization in handling diverse vocabulary, providing a foundation for advanced NLP applications.